

Modeling Divergence Between Community-Perceived and Automatically Calculated MTG Commander Brackets

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Abstract

Magic: The Gathering (MTG) is a collectible card game in which players build decks from thousands of cards and compete in a variety of formats. *Commander* is MTG’s most popular casual multiplayer format, played by four players simultaneously, each with a 100-card singleton deck led by a legendary creature. In this format, players self-report a power level for their decks using a community bracket system (Archidekt, y_1), while automated tools such as EDHPowerLevel independently assign a calculated bracket (y_2) based on objective deck features. This paper studies the divergence between these two evaluations on a dataset of 12,135 public Commander decks (brackets 2–4, $\geq 1,000$ views on Archidekt). We represent each deck in two complementary ways: *Bag of Cards* (BC), which encodes the identity of the cards present, and *Deck Features* (DF), which encodes aggregated structural properties of the deck such as mana curve, card types, combo density, and price. Multiple classifiers are trained on each representation to predict y_1 via nested cross-validation. We further evaluate hard-voting ensembles constructed from out-of-fold predictions without retraining, and compare all model outputs descriptively against y_2 to identify which classifiers converge toward the automated evaluator’s logic. The goal is not to find which source is correct, but to measure and explain the misalignment between two imperfect readings of the same construct.

1 Introduction

Magic: The Gathering (MTG) is a collectible card game first published by Wizards of the Coast in 1993 [1]. Players build decks from a pool of thousands of cards representing spells, creatures, and artifacts, then compete by reducing their opponents’ life points to zero. Cards are played by spending *mana*¹, a resource generated by land cards and spent to cast spells. The game is played across a variety of formats, each with its own rules regarding deck construction, card legality, and number of players.

Commander (also known as EDH, Elder Dragon Highlander) is MTG’s most popular casual multiplayer format, in which four players compete simultaneously. Each deck is built around a legendary creature²—the *commander*—and must contain exactly 100 cards, with at most one copy of each non-basic land card. The social nature of the format means that player experience depends heavily on the power alignment of the decks at the table: when one deck significantly outperforms the others in speed, consistency, or card quality, the game experience degrades—even when all players believe they are playing within a comparable power range. The *Commander Brackets* system was introduced by the format’s official committee as a shared language to communicate expected power level, optimization degree, and the presence of high-impact elements [2].

The Commander Brackets system defines five levels of play, from bracket 1 (casual, low-power, often thematic) to bracket 5 (competitive, cEDH-level). Players are expected to self-assign a bracket to their deck before sitting down to play, and to use this shared vocabulary to negotiate table expectations. However, this self-assignment process is inherently subjective: two players building functionally similar decks may report different brackets based on their local playgroup norms, their personal thresholds for what counts as “powerful,” or simply their familiarity with the format’s power landscape.

Alongside this community self-assessment ecosystem, automated tools have emerged to evaluate Commander decklists algorithmically. These calculators analyze card lists and apply rule-based heuris-

¹*Mana* is the primary resource in MTG. It is produced mainly by *land* cards: each turn, a player may play one land from their hand, and that land can be tapped (turned sideways) to produce one or more mana of a specific color. More expensive cards require more mana to cast, so a player’s available resources grow gradually over the course of a game.

²In MTG, a *legendary creature* is a powerful named character with unique abilities. The Commander rules require each deck to designate one such creature as its *commander*: it begins the game in a separate zone called the *command zone* and can be cast (played) from there, making it consistently accessible throughout the game. The commander’s color identity—determined by the colors of mana symbols in its text and cost—restricts which cards the rest of the deck may include.

tics—counting tutors³, game-winning combinations⁴, high-impact cards, and other structural signals—to assign a bracket without relying on player intent or perception. The divergence between these two sources of bracket information is not merely a technical curiosity: it reveals a gap between what players believe about their decks and what objective structural signals suggest about them.

This gap has practical consequences. A player who underestimates their deck’s power may sit at a table expecting a fair game, only to find that their deck’s structural efficiency exceeds what their bracket label communicated. Conversely, a player who overestimates their deck’s power relative to its actual card list may create social friction by claiming a higher bracket than an automated tool would assign. Understanding the magnitude, direction, and structural correlates of this divergence is therefore relevant both for format design and for the broader question of how community perception relates to measurable deck properties.

We formalize this as a supervised learning and descriptive analysis problem. The two sources of bracket information are treated as two imperfect labels for the same construct:

- y_1 : the bracket assigned by the deck creator on Archidekt [3], a popular deck-building platform with over one million public Commander decklists. This represents *community-perceived* power.
- y_2 : the bracket calculated by EDHPowerLevel [4], an automated tool that applies rule-based criteria to the deck’s card list. This represents *algorithmically-assessed* power.

Crucially, we do not treat either source as ground truth. Neither is objectively correct: y_1 is subject to individual subjectivity, social context, and local playgroup conventions, while y_2 encodes the assumptions built into its rule set and incorporates time-varying signals such as card market prices and EDHREC popularity rankings [5]. Two players can build virtually identical decks and assign different brackets; the same deck can receive different y_2 values on different dates as card prices shift. Our goal is not to arbitrate between these two sources, but to *measure and explain the misalignment* between them.

To study this divergence, we collect a dataset of 12,135 public Commander decks with both labels in $\{2, 3, 4\}$ —brackets 1 and 5 represent structural extremes (purely casual/thematic decks and fully optimized competitive play, respectively) that are sparsely populated, poorly separated from their neighbors by the available features, and subject to high label noise; restricting to the middle range yields a more tractable and better-supported classification problem—and represent each deck in two complementary ways: a *Bag of Cards* (BC) representation that encodes which cards are present, and a *Deck Features* (DF) representation that encodes aggregated structural properties such as mana curve, combo density, card type distribution, and economic signals. We train multiple classifiers on y_1 using nested cross-validation to prevent data leakage, and compare the resulting models descriptively against y_2 to identify which structural signals drive each type of power assessment. This framing allows us to ask not just “how well can we predict the community bracket?” but also “do models that learn community perception converge toward or diverge from algorithmic assessment?”

Research question. *To what extent do Archidekt-assigned Commander brackets diverge from automatically calculated bracket estimates, and which deck characteristics help explain this disagreement?*

Contributions. This paper makes the following contributions:

1. We collect and filter a dataset of 12,950 public Commander decks from Archidekt, retaining 12,135 with both y_1 and y_2 in $\{2, 3, 4\}$ for supervised learning.
2. We perform a direct descriptive analysis of the divergence $\Delta = y_2 - y_1$, characterizing its magnitude, direction, and association with structural deck features.
3. We compare two complementary deck representations—*Bag of Cards* (BC), based on card identity, and *Deck Features* (DF), based on 114 aggregated structural attributes—across 6 classification algorithms using nested cross-validation with no data leakage.

³A *tutor* is a card that searches the player’s library (deck) for any card they need and places it in their hand or into play. The name comes from the iconic card *Demonic Tutor*. Tutors dramatically increase a deck’s consistency by allowing reliable access to key pieces on demand.

⁴A *combo*, short for combination, is a set of two or more cards that, when assembled together, produces a game-ending or overwhelming loop—for example, generating infinite resources or dealing infinite damage. Combos are a central part of high-power play.

4. We evaluate 6 hard-voting ensembles over out-of-fold predictions, and compare all 12 individual models and 6 ensembles against y_2 to identify which predictors converge to the automated evaluator’s logic.
5. We interpret the best BC and DF models to identify which specific cards and structural features most drive community bracket perception.

Paper structure. Section 2 describes the data source, collection, filtering, and labels. Section 3 describes deck representations and the experimental design. Section 4 reports the direct divergence analysis (Section 4.1), spot-checking (Section 4.2), nested CV results (Section 4.3), ensemble evaluation (Section 4.4), and comparison against the calculator (Section 4.5). Section 5 presents model interpretability. Section 6 discusses implications and limitations. Section 7 concludes.

2 Dataset

2.1 Data Source

Data were collected from **Archidekt** [3], a public deck-building platform for MTG that stores user-created Commander decklists along with metadata including card selections, view counts, and a user-assigned bracket. Archidekt exposes a paginated REST API that allows querying public decks by bracket, which we used to collect raw deck payloads. For each deck, we retrieved the full card list and metadata via the deck detail endpoint.

The automated power label y_2 was obtained from **EDHPowerLevel** [4], a web-based calculator that receives a Commander decklist and returns a bracket assignment based on rule-based criteria—including the presence of specific high-impact cards called *Game Changers*⁵, tutor density, combo potential, and card price and popularity signals. Because EDHPowerLevel is a single-page application that computes results in the browser, we automated the interaction using a headless Chromium browser via Playwright, pasting each decklist and extracting the rendered bracket from the DOM.

2.2 Collection and Filtering

Raw collection retrieved 14,421 records from the Archidekt API. After deduplication by deck identifier (1,467 duplicate entries removed) and by card list fingerprint (4 identical lists removed), 12,950 unique decks remained. Decks were retained only if they satisfied all of the following criteria:

- Public deck on Archidekt in the Commander format;
- User-assigned bracket $y_1 \in \{2, 3, 4\}$ — brackets 1 and 5 represent extremes (casual/thematic and competitive cEDH, respectively) with low structural signal density;
- At least 1,000 views on Archidekt, filtering out low-visibility, experimental, or abandoned decklists;
- Exactly 100 cards in the mainboard, as required by Commander rules;
- Only mainboard cards considered — Maybeboard, Sideboard, Tokens, and Extras were excluded.

All 12,950 collected decks already satisfied these filters by construction of the API query. After obtaining y_2 from EDHPowerLevel, decks with $y_2 \in \{1, 5\}$ were excluded from supervised learning (815 decks, 6.3%), leaving a **modeling dataset of 12,135 decks** with both labels in $\{2, 3, 4\}$. These 815 excluded decks are described briefly in Section 4.1 as context for the divergence analysis.

2.3 Labels

Each deck in the modeling dataset has two bracket labels:

- $y_1 \in \{2, 3, 4\}$: the bracket assigned by the deck creator on Archidekt, representing *community-perceived* power. It reflects the player’s own assessment of their deck’s strength, social intent, and play style.

⁵*Game Changers* are cards officially designated by the Commander Rules Committee as capable of winning games or dominating tables on their own, such as *Thassa’s Oracle* (which wins instantly if your library is empty) or *Rhystic Study* (which generates overwhelming card advantage over time). Their presence is the primary factor in distinguishing brackets 3 and 4.

- $y_2 \in \{2, 3, 4\}$: the bracket calculated by EDHPowerLevel, representing *algorithmically-assessed* power. It applies a deterministic rule set to the card list, with signals that include card market price and EDHREC popularity rankings — which may vary over time.

Neither label is treated as ground truth. The supervised models are trained exclusively on y_1 ; y_2 is used only for descriptive comparison after training.

Label distributions in the modeling dataset are shown in Table 1. Both sources are dominated by bracket 3, with class imbalance that motivates the use of macro-F1 as the primary evaluation metric.

Table 1: Label distributions in the modeling dataset (12,135 decks).

Label	Bracket 2	Bracket 3	Bracket 4
y_1 (Archidekt)	2,321 (19.1%)	6,217 (51.2%)	3,597 (29.6%)
y_2 (EDHPowerLevel)	1,911 (15.7%)	6,059 (49.9%)	4,165 (34.3%)

2.4 Temporal Stability of y_2

Because EDHPowerLevel incorporates time-varying signals (card prices and EDHREC popularity), the same deck may receive a different y_2 on different dates. To quantify this instability, we re-queried 90 decks after the initial collection: 87 (96.7%) received the same `commander_bracket`, with the 3 discrepancies concentrated in border decks whose computed power level was within 0.3 points of a bracket boundary. We therefore treat y_2 as a frozen snapshot from the collection date. This limitation is discussed further in Section 6.

3 Methods

3.1 Deck Representations

Each deck in the dataset is represented in two complementary ways, designed to capture different aspects of deck composition.

Bag of Cards (BC). In the BC representation, each deck is encoded as a sparse vector over the vocabulary of all cards observed in the training split of each fold. The value of each dimension corresponds to the count of that card in the deck. Since Commander is a singleton format⁶, most non-basic land cards appear at most once; basic lands may appear with higher counts. The vocabulary is built exclusively from the training partition of each outer fold, and cards not seen during training are silently ignored at test time, preventing any form of data leakage through the feature space. Cards appearing in fewer than 10 training decks are pruned (`min_df` = 10), removing near-zero-variance columns that would inflate dimensionality without contributing signal. This threshold was selected empirically during the spot-checking phase by comparing macro-F1 across `min_df` $\in \{5, 10, 20\}$. The resulting matrix has approximately 11,000 columns.

The BC representation captures whether specific cards are present—staples⁷, combo pieces, and high-impact singletons—and allows models to learn associations between card identity and community bracket assignment.

Deck Features (DF). In the DF representation, each deck is encoded as a dense vector of 114 aggregated structural features extracted from the Archidekt raw JSON. These features are organized into the following groups:

- **Mana base and curve:** land count, basic/non-basic split, mana production by color, and CMC⁸ statistics (mean, median, standard deviation, and counts per CMC bucket from 0 to 6+);

⁶A singleton format requires that each deck contain at most one copy of any given non-basic land card. Basic lands—generic resource-producing cards with no unique text—may be included in any quantity.

⁷A *staple* is an informal term for a card so generically powerful or efficient that it appears in a large fraction of all competitive or semi-competitive decks regardless of their strategy, such as *Sol Ring* (a cheap artifact that produces mana) or *Command Tower* (a land that produces any color of mana in Commander).

⁸*Mana* is the primary resource in MTG, produced mainly by *land* cards. Players play one land per turn and use that land to pay the *mana cost* of spells. The *Converted Mana Cost* (CMC), also called *mana value*, is the total amount of

- **Card types:** counts of creatures, instants, sorceries, artifacts, enchantments, planeswalkers⁹, and non-land permanents;
- **Bracket-relevant flags:** counts of Game Changers¹⁰, tutors¹¹, extra turn effects¹², and mass land denial cards¹³—the four explicit signals that the Commander Brackets system uses to determine bracket thresholds;
- **Combo density:** number of cards involved in atomic combos¹⁴, number of two-card combo pairs, and total combo references;
- **Popularity and price:** EDHREC rank¹⁵ statistics (mean, median, standard deviation), salt score¹⁶ statistics, total and mean deck price;
- **Rarity:** counts of common, uncommon, rare, and mythic cards¹⁷;
- **Keywords:** counts of evergreen keywords¹⁸ (Flying, Trample, Haste, Hexproof, etc.) and total/distinct keyword counts;
- **Card colors and identity:** color count and presence flags for each of the five colors¹⁹;
- **Multiface and layout:** counts of transform²⁰, modal double-faced, split, and adventure cards.

Features derived from user-entered free-text fields (such as Archidekt’s card categories, which are subjective and inconsistently used across decks) were excluded after an audit revealed that 22.8% of decks used no recognizable functional category vocabulary, making such features unreliable as structural signals.

mana required to cast a card, regardless of color. Cards with CMC 0–1 are considered very cheap and fast; cards with CMC 5+ are considered expensive and slow. The distribution of CMC values across a deck is referred to as the *mana curve* and reflects the deck’s expected speed and consistency.

⁹MTG defines several *card types*: *creatures* are units that attack and block; *instants* and *sorceries* are one-time spells; *artifacts* are objects with persistent effects; *enchantments* are ongoing magical effects; *planeswalkers* are powerful allies with activated abilities; *lands* are resource-producing cards. Cards that remain on the battlefield after being played are called *permanents*.

¹⁰*Game Changers* are cards officially designated by the Commander Rules Committee as capable of ending games or dominating tables on their own. Examples include *Thassa’s Oracle* (wins the game immediately under certain conditions) and *Rhystic Study* (generates massive card advantage). The count of Game Changers in a deck is the primary signal used to assign brackets 3 and 4.

¹¹A *tutor* is a card that searches the player’s library (deck) for another specific card and puts it into their hand or into play. Tutors increase deck consistency by allowing players to reliably access key pieces. Examples include *Demonic Tutor* and *Enlightened Tutor*.

¹²Cards that grant a player one or more additional turns beyond the normal one-turn-per-player cycle, such as *Time Warp* or *Temporal Manipulation*, are considered high-impact because they create a significant tempo advantage.

¹³Cards that destroy, prevent the use of, or otherwise deny all players’ lands simultaneously, such as *Armageddon* or *Ravages of War*. These are considered socially contentious in Commander because they can make the game unplayable for opponents.

¹⁴A *combo* in MTG refers to a combination of two or more cards that, when assembled, produces a game-winning or overwhelming effect—for example, dealing infinite damage or generating infinite resources. An *atomic combo* is one where every piece is necessary and sufficient. A *two-card combo* is a particularly efficient variant requiring only two cards.

¹⁵EDHREC [5] is a website that aggregates Commander decklists from the internet and computes statistics for each card based on how often it appears in published decks. The *EDHREC rank* is a popularity ranking: rank 1 means the most widely played card; higher numbers indicate less popular cards. A low EDHREC rank (high popularity) may indicate a generic staple.

¹⁶The *salt score* is a community-sourced metric compiled by EDHREC that measures how much displeasure a card causes other players at the table. High-salt cards are typically those that lock opponents out of playing, cause repeated random effects, or end the game in ways perceived as unfun. A deck with a high average salt score is expected to generate more frustration than a low-salt deck.

¹⁷MTG cards are printed at four rarity levels that broadly correlate with power and availability: *common* (most frequent, generally lower power), *uncommon*, *rare*, and *mythic rare* (least frequent, often the most impactful cards). A deck’s rarity distribution is a proxy for its overall power ceiling and monetary cost.

¹⁸*Keywords* in MTG are shorthand labels for recurring abilities. *Evergreen* keywords appear across many sets and are always available: examples include *Flying* (can only be blocked by creatures with Flying or Reach), *Trample* (excess damage carries over to the player), *Haste* (can attack immediately), *Hexproof* (cannot be targeted by opponents), and *Deathtouch* (destroys any creature it damages).

¹⁹MTG uses five colors of mana and magic: *White* (W), *Blue* (U), *Black* (B), *Red* (R), and *Green* (G). Each color has a distinct playstyle: White emphasizes order and removal; Blue favors control and card draw; Black uses sacrifice and tutors; Red relies on speed and direct damage; Green ramps mana and uses large creatures. A deck’s color identity constrains which cards it may include, as determined by its commander.

²⁰Some MTG cards have two faces and can *transform* between them under certain conditions—for example, a werewolf that transforms into a more powerful form at night. *Modal double-faced cards* let the player choose which face to play. *Split cards* have two independent spells on one card. *Adventure* cards combine a creature with an instant or sorcery. These layouts can affect deck construction and strategic flexibility.

3.2 Preprocessing Pipeline

All preprocessing transformations are applied *within each cross-validation fold*, fit exclusively on the training partition. No global statistics are computed over the full dataset before splitting. Specifically:

- **Imputation:** median imputation is applied to EDHREC rank and salt score features, which may be missing for rare or recently released cards. The median is computed on the training fold only.
- **Winsorization:** the total deck price (`price_total`) is winsorized at the 99th percentile of the training fold, capping extreme outliers (some decks exceed \$50,000) without distorting the bulk of the distribution.
- **Scaling:** `StandardScaler` is applied to DF features for scale-sensitive algorithms (Logistic Regression, LinearSVC, and KNN). Tree-based models (Decision Tree, Random Forest, Gradient Boosting) and Gaussian Naive Bayes receive unscaled features.
- **Leakage prevention:** all fields derived from the EDHPowerLevel output (y_2 , power level score, efficiency, tipping point, and all prefixed `ep1_` columns) are explicitly blocked from entering any feature matrix. The only training target is y_1 .

3.3 Candidate Algorithms

We evaluated seven classification algorithms chosen to cover diverse inductive biases:

Table 2: Candidate algorithms and their inductive biases.

Algorithm	sklearn Class	Inductive Bias
Decision Tree	<code>DecisionTreeClassifier</code>	Tree partitioning
Random Forest	<code>RandomForestClassifier</code>	Bagging
Gradient Boosting	<code>HistGradientBoostingClassifier</code>	Boosting
Naive Bayes	<code>MultinomialNB</code> (BC) / <code>GaussianNB</code> (DF)	Probabilistic
Logistic Regression	<code>LogisticRegression</code>	Linear parametric
Linear SVC	<code>LinearSVC</code>	Linear margin
K-Nearest Neighbors	<code>KNeighborsClassifier</code>	Distance-based

Kernel SVMs (RBF and polynomial) were excluded because they do not scale to the BC representation’s high-dimensional sparse space ($\sim 12,000$ decks $\times \sim 11,000$ card dimensions), violating the requirement that all candidate algorithms must be viable in both representations. Non-linear capacity is instead provided by Gradient Boosting and KNN.

3.4 Spot-Checking

Before investing in costly hyperparameter optimization, we performed a spot-checking phase to filter the algorithm pool. Each of the seven candidates was evaluated on both representations with default hyperparameters, using $N = 5$ stratified hold-out splits (80/20) with seeds $\{1, 2, 3, 4, 5\}$. We report the mean and standard deviation of macro-F1 across the five seeds for each (algorithm, representation) combination. The top-5 algorithms per representation were selected independently by mean macro-F1, and their union $A_{\text{union}} = A_{\text{BC}} \cup A_{\text{DF}}$ defines the set of algorithms that proceed to nested cross-validation. This resulted in 6 algorithms in the union and 12 models to train (6×2 representations).

3.5 Nested Cross-Validation

All selected models are evaluated via **nested cross-validation** to obtain unbiased estimates of generalization performance while simultaneously performing hyperparameter selection:

- **Outer loop:** `StratifiedKFold` with 5 splits and 3 independent repeats (seeds 1, 2, 3), yielding 15 outer evaluations per model. The outer loop estimates final generalization performance.
- **Inner loop:** `StratifiedKFold` with 3 splits (seed = outer seed + 100), used for grid search over hyperparameters. The inner loop selects the best configuration for each outer fold without touching the outer test set.

All models use identical fold splits, enabling paired statistical comparisons. Fold indices are deterministically derived from the label vector and random seeds at runtime, ensuring cross-machine reproducibility given the same dataset and dependency versions.

Each algorithm was evaluated over a compact grid of 24 configurations, balancing coverage of the most impactful hyperparameters against computational cost. Table 3 summarizes the hyperparameter grids.

Table 3: Hyperparameter grids for nested CV (24 configurations per algorithm).

Algorithm	Parameter	Values
Decision Tree	<code>max_depth</code>	{None, 10, 20}
	<code>min_samples_leaf</code>	{1, 5}
	<code>ccp_alpha</code>	{0, 0.005}
	<code>class_weight</code>	{None, balanced}
Random Forest	<code>n_estimators</code>	{100, 250, 500}
	<code>max_features</code>	{sqrt, log2}
	<code>min_samples_leaf</code>	{1, 2}
	<code>class_weight</code>	{None, balanced}
Gradient Boosting	<code>max_iter</code>	{200, 500}
	<code>learning_rate</code>	{0.05, 0.1, 0.2}
	<code>max_leaf_nodes</code>	{15, 31}
	<code>class_weight</code>	{None, balanced}
Naive Bayes (BC)	<code>alpha</code>	12 log-spaced values in $[10^{-3}, 100]$
	<code>fit_prior</code>	{True, False}
Naive Bayes (DF)	<code>var_smoothing</code>	24 log-spaced values in $[10^{-12}, 10^{-3}]$
Logistic Regression	<code>C</code>	{0.001, 0.1, 1, 100}
	<code>class_weight</code>	{None, balanced}
	<code>l1_ratio</code>	{0, 0.5, 1} (ElasticNet penalty)
Linear SVC	<code>C</code>	{0.001, 0.01, 0.1, 1, 10, 100}
	<code>class_weight</code>	{None, balanced}
	<code>penalty</code>	{L1, L2}

The inner selection criterion is macro-F1. The outer loop reports macro-F1, accuracy, precision (macro), recall (macro), and confusion matrices. We report the mean and standard deviation of each metric across the 15 outer evaluations.

3.6 Voting Ensembles

After nested CV, we construct six hard-voting ensembles from the out-of-fold predictions of the individual models, without any additional training:

- `voting_top3_BC`: top-3 BC models by mean macro-F1;
- `voting_top5_BC`: top-5 BC models;
- `voting_top3_DF`: top-3 DF models;
- `voting_top5_DF`: top-5 DF models;
- `voting_top3_BC_DF`: top-3 BC + top-3 DF models (6 members);
- `voting_all`: all individual models.

For each ensemble, the predicted class for a given deck is the plurality vote across member predictions. Ties are broken by selecting the class whose voting members have the highest mean macro-F1; if still tied, the numerically smallest class label is chosen, ensuring deterministic behavior. Performance is evaluated using the same fold partitions as the individual models, and mean $\pm$ standard deviation over the 15 outer folds is reported.

3.7 Comparison Against the Calculator

Since models are trained exclusively on y_1 , the out-of-fold predictions $\hat{y}_1$ are compared *descriptively* against y_2 (the automated calculator’s bracket) after training is complete. This comparison does not involve any retraining or new targets. For each individual model and each ensemble, we compute: exact agreement between $\hat{y}_1$ and y_2 , agreement within ± 1 , mean absolute deviation $|\hat{y}_1 - y_2|$, macro-F1 treating y_2 as the reference label, and the confusion matrix $\hat{y}_1 \times y_2$. This analysis reveals which models converge toward the automated evaluator’s logic and which capture community-specific patterns not reflected in the rule-based calculator.

3.8 Evaluation Metric

The primary metric for model selection and comparison is **macro-F1**, which averages the F1 score of each class with equal weight. This choice is appropriate given the class imbalance in y_1 (bracket 3 accounts for 51.2% of decks), where accuracy alone could mask poor performance on minority classes. Accuracy, macro-precision, macro-recall, and per-class confusion matrices are reported as complementary metrics.

4 Results

4.1 Direct Divergence Analysis

We first characterize the raw disagreement between the two labels before training any model. This includes:

- Exact agreement rate and within- ± 1 agreement between y_1 and y_2 ;
- Signed distribution of $\Delta = y_2 - y_1$ and its directional asymmetry;
- Cross-tabulation $y_1 \times y_2$;
- Modulation of $|\Delta|$ by structural deck features (Game Changer count, tutor density, price quintile, color count).

4.2 Spot-Checking

We report the mean macro-F1 over five stratified hold-out seeds for all seven candidate algorithms on both representations, including:

- Full results table (algorithm $\times$ representation) with mean $\pm$ std;
- Selected `bc_min_df` value and justification;
- Top-5 ranking per representation and union A_{union} .

4.3 Nested Cross-Validation

We report nested CV performance (15 outer folds: 5 splits $\times$ 3 repeats) for all 12 models (6 algorithms $\times$ 2 representations), covering:

- Macro-F1, accuracy, macro-precision and macro-recall per model;
- DF vs. BC representation gap analysis;
- Friedman test across all models and post-hoc Nemenyi pairwise comparisons;
- Paired Wilcoxon significance tests for key model pairs.

4.4 Voting Ensembles

We report the six hard-voting ensembles evaluated on the same 15 outer folds, covering:

- Macro-F1 and accuracy per ensemble;
- Comparison against the best individual model;
- Effect of mixing BC and DF members.

4.5 Comparison Against the Calculator

We compare out-of-fold predictions $\hat{y}_1$ from all models and ensembles descriptively against y_2 , covering:

- Exact agreement rate $\hat{y}_1 = y_2$ and within- ± 1 agreement per model;
- Mean absolute deviation $|\hat{y}_1 - y_2|$;
- Macro-F1 treating y_2 as reference label;
- Discussion of which representations converge toward or diverge from the calculator’s logic.

5 Model Interpretability

6 Discussion

7 Conclusion

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